REPORTT-Retail Sales Analytics Dashboard

Project Overview

The Retail Sales Analytics Dashboard is an end-to-end business intelligence project developed to analyze retail sales data and generate actionable insights. The project uses Python for data preprocessing and analysis and Power BI for interactive dashboard development.

Architecture

Backend

The backend layer is responsible for data processing and analytics.

Technologies Used

- Python
- Pandas
- SQL
- CSV Dataset

Backend Functionalities

- Data ingestion and preprocessing
- Missing value and duplicate detection
- Date transformation and formatting
- Sales and profit calculations
- Regional and category-wise analysis
- Customer segmentation analysis
- Top product identification
- Data export for dashboard visualization

Frontend

The frontend layer provides interactive visualization and reporting capabilities.

Technologies Used

- Power BI

- DAX Measures

Frontend Functionalities

- KPI Cards for Sales, Profit, Orders, and Average Sales
- Sales Trend Analysis
- Sales by Region Visualization
- Profit by Category Analysis
- Customer Segment Distribution
- Top Products Analysis
- Interactive Filters and Slicers
- Dynamic Dashboard Reporting

Dataset Information

- Dataset: Sample Superstore Dataset
- Total Records: 9,994
- Total Columns: 21

Data Preprocessing

- Imported dataset using Pandas.
- Checked and validated data types.
- Verified missing values and duplicate records.
- Converted date columns into datetime format.
- Generated descriptive statistics.
- Exported cleaned dataset for reporting.

Dashboard Components

KPI Metrics

- Total Sales: ₹2.30 Million
- Total Profit: ₹286 Thousand
- Total Orders: 5K

- Average Sales: ₹229.86

Visualizations

- Sales Trend Over Time
- Sales by Category
- Profit by Category
- Sales by Region
- Sales by Customer Segment
- Top 10 Products by Sales

Filters

- Region
- Category
- Customer Segment

Key Insights

- The West region generated the highest revenue.
- Technology category delivered the maximum profit.
- Consumer segment contributed more than 50% of overall sales.
- Top-performing products were identified using sales metrics.
- Interactive reporting enabled efficient business decision-making.

Conclusion

The Retail Sales Analytics Dashboard demonstrates the complete data analytics workflow, from data preprocessing and analysis to visualization and business intelligence reporting. The project showcases proficiency in Python, SQL, Power BI, and analytical problem-solving.

Skills Demonstrated

- Python Programming
- Pandas
- SQL

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Data Visualization
- Power BI
- DAX
- Business Intelligence
- Dashboard Development